COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

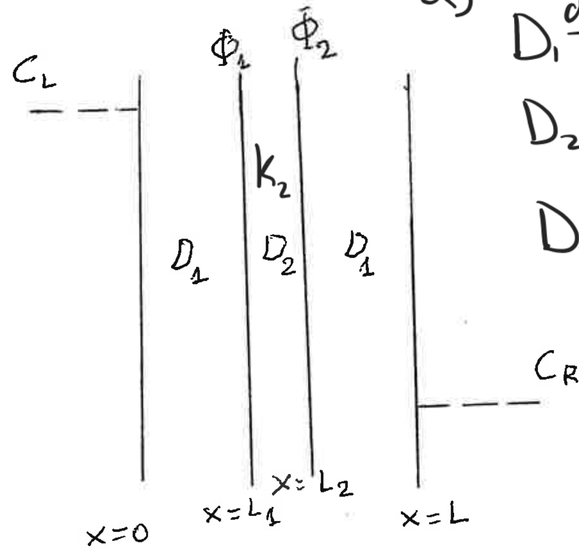
Exam #3

Duration: 70 minutes* --- *November 25, 2025

All Problems must be done

Problem 1: (30 points)
Consider steady state diffusion of the species A through three media arranged as shown in the attached diagram. Assume that diffusion is one-dimensional. At x = 0, $C_{A1} = C_L$, and at x = L, $C_{A3} = C_R$. In medium 2, there is a zeroth order chemical reaction through which the species A is consumed. Develop an expression for the concentration profiles across the three media.

(Bonus 10 points) Develop an expression for the steady state flux across the three media.



a)

$$D_1 \frac{d^2 C_A}{dx^2} = 0$$

$$D_2 \frac{d^2 C_A}{dx^2} - K_A = 0$$

$$D_3 \frac{d^2 C_A}{dx^2} = 0$$

BC's

$C_a(x=0) = C_L$ ✓

$C_a(x=L) = C_R$ ✓

$-D_1\left(\frac{dC_{A_1}}{dx}\right) = -D_2\left(\frac{dC_{A_2}}{dx}\right)\Big|_{x=L_1}$

$-D_2\left(\frac{dC_{A_2}}{dx}\right) = -D_3\left(\frac{dC_{A_3}}{dx}\right)\Big|_{x=L_2}$

$\Phi_1 = \frac{C_{A_2}}{C_{A_1}}\Big|_{x=L_1}$ $\quad \Phi_2 = \frac{C_{A_3}}{C_{A_2}}\Big|_{x=L_2}$

$$C_{A_1} = Bx + D$$

$$C_{A_2} = \frac{k_A}{2D_2}x^2 + Ex + F$$

$$C_{A_3} = Gx + H$$

$C_L = 0 + \boxed{D = C_L}$

$C_R = 0 + \boxed{H = C_R}$

$-D_1(B) = -D_2\left(\frac{k_A}{D_2}x + E\right)\Big|_{x=L_1}$

$-D_2\left(\frac{k_A}{D_2}x + E\right) = -D_3(G)\Big|_{x=L_2}$

$$\Phi_1 = \frac{\frac{k_A}{2D_2}x^2 + Ex + F}{Bx + D}\Bigg|_{x=L_1} \rightarrow \frac{\frac{k_A}{2D_2}(L_1)^2 + E(L_1) + F}{Bx + C_L}$$

$$\Phi_2 = \frac{Gx + H}{\frac{k_A}{2D_2}x^2 + Ex + F}\Bigg|_{x=L_2} \rightarrow \frac{G(L_2) + C_R}{\frac{k_A}{2D_2}(L_2)^2 + E(L_2) + F}$$

$$-D_1(B) = -D_2\left(\frac{k_A}{D_2}(L_1)+E\right)$$

$$-D_2\left(\frac{k_A}{D_2}(L_2)+E\right) = -D_3(G)$$

$$B = \frac{D_2}{D_1}\left(\frac{k_A}{D_2}L_1+E\right)$$

$$G = \frac{D_2}{D_3}\left(\frac{k_A}{D_2}L_2+E\right)$$

$$\frac{\frac{k_A}{2D_2}(L_1)^2+E(L_1)+F}{BL_1+C_L} = \phi_1$$

$$\frac{G(L_2)+C_R}{\frac{k_A}{2D_2}(L_2)^2+E(L_2)+F} = \phi_2$$

$$\frac{\frac{k_A}{2D_2}L_1^2+EL_1+F}{\frac{D_2}{D_1}\left(\frac{k_A}{D_2}L_1+E\right)L_1+C_L} = \phi_1$$

$$\frac{\frac{D_2}{D_3}\left(\frac{k_A}{D_2}L_2+E\right)L_2+C_R}{\frac{k_A}{2D_2}(L_2)^2+E(L_2)+F} = \phi_2$$

$$\frac{k_A}{2D_2}L_1^2+EL_1+F = \phi_1\left(\frac{D_2}{D_1}\left(\frac{k_A}{D_2}L_1+E\right)L_1+C_L\right)$$

$$F = \phi_1\left(\frac{D_2}{D_1}\left(\frac{k_A}{D_2}L_1+E\right)L_1+C_L\right)-\frac{k_A}{2D_2}L_1^2-EL_1$$

so

$$\phi_1\left(\frac{k_A}{D_1}L_1^2+E\frac{L_1D_2}{D_1}+C_L\right)-\frac{k_A}{2D_2}L_1^2-EL_1$$

$$\frac{k_A\phi_1}{D_1}L_1^2+E\frac{L_1D_2\phi_1}{D_1}+C_L\phi_1-\frac{k_A}{2D_2}L_1^2-EL_1$$

$$F = E\left(\frac{L_1D_2\phi_1}{D_1}-L_1\right)+\frac{k_A\phi_1}{D_1}L_1^2+C_L\phi_1-\frac{k_A}{2D_2}L_1^2$$

$$\Psi = \frac{L_1D_2\phi_1}{D_1}-L_1 \qquad g = \frac{k_A\phi_1}{D_1}L_1^2+C_L\phi_1-\frac{k_A}{2D_2}L_1^2$$

$$\frac{\frac{D_2}{D_3}\left(\frac{k_A}{D_2}L_2+E\right)L_2+C_R}{\frac{k_A}{2D_2}L_2^2+EL_2+\phi_1\left(\frac{D_2}{D_1}\left(\frac{k_A}{D_2}L_1+E\right)L_1+C_L\right)-\frac{k_A}{2D_2}L_1^2-EL_1} = \phi_2 \to \frac{\frac{D_2}{D_3}\left(\frac{k_A}{D_2}L_2^2+EL_2\right)+C_R}{\frac{k_A}{2D_2}L_2^2+EL_2+\phi_1\left(\frac{D_2}{D_1}\left(\frac{k_A}{D_2}L_1^2+EL_1\right)+C_L\right)-\frac{k_A}{2D_2}L_1^2-EL_1}$$

$$\phi_2 = \frac{E\frac{L_2D_2}{D_3}+\frac{k_A}{D_3}L_2^2+C_R}{EL_2+E\frac{L_1D_2\phi_1}{D_1}-EL_1+\frac{k_A}{2D_2}L_2^2+\frac{\phi_1k_A}{D_1}L_1^2+L_L\phi_1-\frac{k_A}{2D_2}L_1^2} \to \frac{E\frac{L_2D_2}{D_3}+\frac{k_A}{D_3}L_2^2+C_R}{E\left(L_2+\frac{L_1D_2\phi_1}{D_1}-L_1\right)+\frac{k_A}{2D_2}L_2^2+\frac{\phi_1k_A}{D_1}L_1^2+L_L\phi_1-\frac{k_A}{2D_2}L_1^2}$$

$$\alpha = \frac{k_A}{D_3}L_2^2+C_R \qquad \beta = \frac{L_2D_2}{D_3} \qquad \mu = L_2+\frac{L_1D_2\phi_1}{D_1}-L_1 \qquad \sigma = \frac{k_A}{2D_2}L_2^2+\frac{\phi_1k_A}{D_1}L_1^2+C_L\phi_1-\frac{k_A}{2D_2}L_1^2$$

so

$$\phi_2 = \frac{E\beta+\alpha}{E\mu+\sigma} \to E\mu+\sigma = \frac{E\beta+\alpha}{\phi_2} \to E\mu\phi_2+\sigma\phi_2-E\beta-\alpha=0 \to E(\mu\phi_2-\beta)+\sigma\phi_2-\alpha=0$$

$$E = \frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta} \longrightarrow F = \frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\Psi+g$$

$$B = \frac{k_A}{D_1}L_1+E\frac{D_2}{D_1} \to \frac{k_A}{D_1}L_1+\frac{D_2}{D_1}\left(\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\right) \qquad G = \frac{k_A}{D_3}L_2+E\frac{D_2}{D_3} \to \frac{k_A}{D_3}L_2+\frac{D_2}{D_3}\left(\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\right)$$

$$C_{A_1} = Bx+D \longrightarrow \left(\frac{k_A}{D_1}L_1+\frac{D_2}{D_1}\left(\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\right)\right)x+C_L$$

$$C_{A_2} = \frac{k_A}{2D_2}x^2+Ex+F \to \frac{k_A}{2D_2}x^2+\left(\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\right)x+\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\Psi+g$$

$$C_{A_3} = Gx+H \to \left(\frac{k_A}{D_3}L_2+\frac{D_2}{D_3}\left(\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\right)\right)x+C_R$$

b) $$N_{A_1}'' = -D_1\left(\frac{k_A}{D_1}L_1+\frac{D_2}{D_1}\left(\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\right)\right) \to -k_AL_1-D_2\left(\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\right)$$

$$N_{A_2}'' = -D_2\left(\frac{k_A}{D_2}x+E\right) \to -k_Ax-\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}$$

$$N_{A_3}'' = -D_3\left(\frac{k_A}{D_3}L_2+\frac{D_2}{D_3}\left(\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\right)\right) \to -k_AL_2-D_2\left(\frac{\alpha-\sigma\phi_2}{\mu\phi_2-\beta}\right)$$

Problem 2: (70 points)

Consider the reversible conversion of species A into B through the stoichiometric reactions A $\underset{k_{-1}\ K_2}{\overset{k_1\ K_1}{\rightleftharpoons}}$ B in a stationary liquid film of thickness L. Determine C_A and C_B throughout the film, if

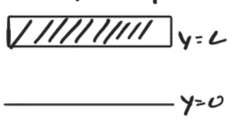
a) (**35 Points**) the surface at y = L is assumed to be impermeable and unreactive, and the concentrations of A and B at y = 0 are specified as C_{Ao} and C_{Bo}. The forward and reverse homogeneous reactions follow zeroth order kinetics

b) (**35 Points**) the surface at y = L is assumed to be impermeable and unreactive, and the concentrations of A and B at y = 0 are specified as C_{Ao} and C_{Bo}. The forward and reverse homogeneous reactions follow zeroth and first order kinetics, respectively

a)

$$D\frac{d^2C_A}{dy^2}\underbrace{-K_a+K_b}_{K_R}\rightarrow D\frac{d^2C_A}{dy^2}+K_R=0\rightarrow\frac{d^2C_A}{dy^2}=-\frac{K_R}{D}\rightarrow C_A=-\frac{K_R}{2D}y^2+Ey+F$$

$$D\frac{d^2C_B}{dy^2}\underbrace{-K_b+K_a}_{K_R}\rightarrow D\frac{d^2C_B}{dy^2}-K_R=0\rightarrow C_B=\frac{K_R}{2D}y^2+Gy+H$$



$$C_A(y=0)=C_{Ao}\quad -D\left(\frac{dC_A}{dy}\right)\Big|_{y=L}=0\qquad C_{Ao}=0+0+F\rightarrow F=C_{Ao}$$

$$C_B(y=0)=C_{Bo}\quad -D\left(\frac{dC_B}{dy}\right)\Big|_{y=L}=0\qquad C_{Bo}=0+0+H\rightarrow H=C_{Bo}$$

$$-D\left(-\frac{K_R}{D}y+E\right)=0\rightarrow E=\frac{K_R}{D}L\qquad F=-\frac{K_R}{D}L$$

$$C_A=-\frac{K_R}{2D}y^2+\frac{K_R}{D}Ly+C_{Ao}\qquad C_B=\frac{K_R}{2D}y^2-\frac{K_R}{D}Ly+C_{Bo}$$

b)

$$D\frac{d^2C_A}{dy^2}-K_A+k_BC_B=0$$

$$D\frac{d^2C_B}{dy^2}+K_A-K_BC_B=0$$

$$D\frac{d^2C_A}{dy^2}=K_A-K_BC_B\rightarrow\frac{d^2C_A}{dy^2}=\frac{K_A-K_BC_B}{D}\rightarrow\frac{dC_A}{dy}=\frac{K_A-K_BC_B}{D}y+E\rightarrow C_A=\frac{K_A-K_BC_B}{2D}y^2+Ey+F$$

$$D\frac{d^2C_B}{dy^2}+K_A-K_BC_B=0$$

will be λ

homogenous: $D\frac{d^2C_B}{dy^2}-K_BC_B=0\rightarrow Dr^2-K_B=0\rightarrow r=\pm\sqrt{\frac{K_B}{D}}$ so $C_B=Ge^{\lambda y}+He^{-\lambda y}$

Particular: $D\frac{d^2C_B}{dy^2}+K_A-K_BC_B=0\rightarrow C_B=-\frac{K_A}{K_B}$ so Characterstic: $C_B=Ge^{\lambda y}+He^{-\lambda y}-\frac{K_A}{K_B}$

$C_A=\frac{K_A-K_BC_B}{2D}y^2+Ey+F\rightarrow C_{Ao}=F,\ -D\left(\frac{dC_A}{dy}\right)=0\rightarrow E=-\frac{K_A-K_BC_B}{D}\rightarrow C_A=\frac{K_A-K_BC_B}{2D}y^2-\frac{K_A-K_BC_B}{D}y+C_{Ao}$

$C_B=Ge^{\lambda y}+He^{-\lambda y}-\frac{K_A}{K_B}$

$C_{Bo}=G+H-\frac{K_A}{K_B}$

$0=\lambda Ge^{\lambda L}-\lambda He^{-\lambda L}\rightarrow 0=(Ge^{\lambda L}-He^{-\lambda L})\lambda\rightarrow 0=Ge^{\lambda L}-He^{-\lambda L}$